

Erdős Problem #1112

EP-1112

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $1 \leq d_1 < d_2$ and $k \geq 3$. Does there exist an integer r such that if $B = \{b_1 < \dots\}$ is a lacunary sequence of positive integers with $b_{i+1} \geq rb_i$ then there exists a sequence of positive integers $A = \{a_1 < \dots\}$ such that $d_1 \leq a_{i+1} - a_i \leq d_2$ for all $i \geq 1$ and $(kA) \cap B = \emptyset$, where kA is the k -fold sumset?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>